MATH 350-2 Advanced Calculus

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Outline

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Riemann-Stieltjes sums

Notation: given a real-valued function α on $[a, b]$

$$\Delta\alpha_k(P) = \alpha(x_k) - \alpha(x_{k-1}), \quad k = 1, 2, \dots, n.$$

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Definition

Let f and α be real-valued functions on $[a, b]$, and let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. A **Riemann-Stieltjes sum** is a sum of the form

$$S(P, f, \alpha, \{t_k\}) = \sum_{k=1}^{n} f(t_k)\Delta\alpha_k(P),$$

for some set of sample points $\{t_k\}$ of P.

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$$|S(P, f, \alpha, \{t_k\}) - A| < \epsilon,$$

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Notation: in this case A is denote by $\int_a^b f d\alpha$ or $\int_a^b f(x) d\alpha(x)$

Challenge!

Problem

Suppose that a, b are real numbers with $a < b$ and let α be a real-valued function on $[a, b]$.
Prove that a constant function $f(x) = c$ is integrable with respect to α on $[a, b]$ and

$$\int_a^b f d\alpha = c\alpha(b) - c\alpha(a).$$

Challenge!

Problem

Suppose that a, b are real numbers with $a < b$ and let α be a constant function on $a < b$.
If $f(x)$ is a function on $[a, b]$, show that $f(x)$ is integrable with respect to α on $[a, b]$.

$$\text{What is } \int_a^b f d\alpha?$$

Challenge!

Problem

Suppose that a, b, c are real numbers with $a < c < b$ and let α be the step function

$$\alpha(x) = \begin{cases} 0, & x < c \\ 1, & x \geq c \end{cases}$$

Show that if $f(x)$ is a real function on $[a, b]$ which is continuous at c, then f is integrable with respect to α on $[a, b]$.

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Linearity of integration

Theorem

Suppose that f and g are Riemann integrable with respect to α on $[a, b]$.

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Suppose that f and g are Riemann integrable with respect to α on $[a, b]$.
Then for any $c_1, c_2 \in \mathbb{R}$, $c_1 f + c_2 g$ is Riemann integrable with respect to α and

$$\int_a^b (c_1 f + c_2 g) d\alpha = c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha.$$

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Suppose that f is Riemann integrable with respect to both α and β on $[a, b]$.
Then for any $c_1, c_2 \in \mathbb{R}$, f is Riemann integrable with respect to $c_1\alpha + c_2\beta$ and

$$\int_a^b f d(c_1\alpha + c_2\beta) = c_1 \int_a^b f d\alpha + c_2 \int_a^b f d\beta.$$

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$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

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Suppose that f is a real-valued function on $[a, b]$ and that $c \in (a, b)$. If $\int_a^c f d\alpha$ and $\int_b^c f d\alpha$ exist, then $\int_a^c f d\alpha$ exists and

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Conversely, if $\int_a^c f d\alpha$ exists then $\int_a^c f d\alpha$ and $\int_b^c f d\alpha$ exist and

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

Integration by parts

The integrand and integrator play dual roles:

Theorem

Integration by parts Let f be integrable with respect to α on $[a, b]$.
Then α is integrable with respect to f on $[a, b]$ and

$$\int_a^b f d\alpha = f(b)\alpha(b) - f(a)\alpha(a) - \int_a^b \alpha df.$$

Relation to Riemann integrals

Theorem

Let f be integrable with respect to α on $[a, b]$.
If α is continuous on $[a, b]$ and continuously differentiable on (a, b), then $f(x)\alpha'(x)$ is integrable with respect to x on $[a, b]$ and

$$\int_a^b f d\alpha = \int_a^b f(x)\alpha'(x) dx.$$

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Concrete Example

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Take a partition $P = \{x_0, \ldots, x_n\}$ of $[0, 1]$, sample points $\{t_k\}$:

$$S(P, f, \alpha, \{t_k\} = \sum_{k=1}^{n} f(t_k)\Delta\alpha_k(P) = \sum_{k=1}^{n} t_k(x_k - x_{k-1}).$$

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Since x is monotone increasing:

$$\sum_{k=1}^{n} x_{k-1}(x_k - x_{k-1}) \leq S(P, f, \alpha, \{t_k\}) \leq \sum_{k=1}^{n} x_k(x_k - x_{k-1}).$$

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Where do you go from here?

Stieltjes sums

Let f and α be real-valued functions on $[a, b]$, and let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$ and define

$$m_k(f) = \inf\{f(x) : x_{k-1} \le x \le x_k\}, \ \ M_k(f) = \sup\{f(x) : x_{k-1} \le x \le x_k\}.$$

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The **upper Stieltjes sum** of f with respect to α on $[a, b]$ is:

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The **upper Stieltjes sum** of f with respect to α on $[a, b]$ is:

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and the **lower Stieltjes sum** of f with respect to α on $[a, b]$ is:

$$\underline{S}(P, f, \alpha) = \sum_{k=1}^{n} m_k(f)\Delta\alpha_k(P)$$

Stieltjes sums

These sums refine predictably!

Lemma

Let f and α be real-valued functions on $[a, b]$, with α monotone increasing.

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Lemma

Let f and α be real-valued functions on $[a,b]$, with α monotone increasing. If P is a partition of $[a,b]$ and P' is a refinement of P, then

$$\underline{S}(P,f,\alpha) \le \underline{S}(P',f,\alpha) \le \overline{S}(P',f,\alpha) \le \overline{S}(P,f,\alpha)$$

Stieltjes integrals

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Let f and α be real-valued functions on $[a, b]$, with α monotone increasing. The **upper Stieltjes integral** of f with respect to α on $[a, b]$ is:

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The **lower Stieltjes integral** of f with respect to α on $[a, b]$ is:

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Riemann's criterion

Definition

Let f and α be real-valued functions on $[a, b]$.
Then f **satisfies Riemann's criterion** with respect to α on $[a, b]$ if for all $\epsilon > 0$ there exists a partition P_ϵ such that

$$0 \leq \overline{S}(P, f, \alpha) - \underline{S}(P, f, \alpha) < \epsilon.$$

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(a) f is Riemann integrable with respect to α on $[a, b]$
(b) f satisfies Riemann's criterion with respect to α on $[a, b]$
(c) $\underline{I}(f, \alpha) = \overline{I}(f, \alpha)$